

7 Discussion

7.1 Main findings

On a held-out IDRiD test set that played no role in any modelling decision, NITSCI reached a five-class accuracy of **[acc5]**% with a QWK of **[qwk5]**, graded **[within1]**% of images within one level of the reference, and detected referable DR with an AUC of **[refauc]** at a sensitivity of **[refsens]**% and a specificity of **[refspec]**%. *[AUTHOR: Summarise in two or three sentences, using only claims supported by Tables 4–8: (1) whether the full model improves on the plain baseline beyond the bootstrap interval; (2) which ablations had the largest effect; (3) whether the ordinal head and threshold fitting reduced large errors, visible in QWK and within-one-grade accuracy.]*

7.2 Interpretation of the components

Cross-dataset pretraining. APTOS-2019 differs from IDRiD in cameras, image quality and population, but shares the grading scale and the visual vocabulary of DR lesions. Pretraining on it exposes the model to roughly eight times more graded fundus images before it sees IDRiD, which is consistent with the general observation that in-domain data narrows the gap left by ImageNet pretraining [57]. *[AUTHOR: Report the size of the “w/o APTOS” effect from Table 7 and whether it is the largest single factor.]*

Lesion expert, cross-attention and gate. The three architectural components address the same problem from different angles: the lesion map concentrates the representation on a few informative locations, cross-attention lets lesion tokens be interpreted in their retinal context, and the gate allows the balance between global and lesion evidence to vary from image to image. A plausible expectation is that the gate favours lesion evidence for borderline images between grades 0, 1 and 2, where the decision depends on a few lesions, and global context for advanced disease, where lesions are widespread. *[AUTHOR: Check this with the gate values exported for the test images, or remove the sentence. Report which of the three ablations changed accuracy or QWK beyond the confidence interval, and say plainly if a component had no measurable effect at this test-set size.]*

Ordinal head and decision rule. Combining the expected grade of the classifier with a regression output produces a score that changes smoothly with disease severity, so errors tend to fall on adjacent grades. Fitting the cut-points on out-of-fold predictions adapts them to the class balance of IDRiD without touching the test set. The comparison of decision rules in Table 8 shows how much of the final accuracy is due to the rule rather than to the network. *[AUTHOR: Comment on whether the OOF ranking of the rules matched their test ranking; if not, this indicates that the threshold fit partly reflects noise in a small OOF set.]*

Ensembling and TTA. Averaging 20 predictions per image reduces variance, which matters most for the under-represented grades. The difference between a single fold model and the ensemble (Table 8) also quantifies a practical trade-off: a single model without TTA is twenty times cheaper at inference.

7.3 Comparison with published results

Direct numerical comparison with the IDRiD grading challenge [13] is not possible from this study, because the challenge uses an official train/test partition whereas we use a stratified random hold-out of the Kaggle copy. Numbers reported on different splits, with different pre-processing and sometimes with thresholds tuned on the test set, are not comparable, and placing them in the same table would mislead the reader. We therefore compare against a plain baseline and ablations

trained under identical conditions (Table 7). *[AUTHOR: Either (a) re-run the pipeline on the official IDRiD grading partition and add a table comparing with the challenge results reported in Porwal et al. [13], copying the numbers from that paper; or (b) keep this paragraph.]*

7.4 Interpretability

The lesion-attention map and Grad-CAM [77] offer a view of where the model looks, which can help clinicians to spot obvious failure modes. They should not be over-interpreted: saliency maps can be insensitive to the model parameters [78], and post-hoc explanations of black-box models are not a substitute for validated, inherently interpretable evidence in high-stakes decisions [79]. Because our lesion map is learned without lesion masks, we treat it as a qualitative aid only; validating it against the IDRiD lesion annotations is part of future work.

7.5 Limitations

- **Small test set.** With **[ntest]** test images, confidence intervals are wide, and per-grade metrics for rare grades rest on few images. Differences of a few percentage points between configurations may not be statistically detectable.
- **Single target dataset.** All results come from IDRiD, collected at one centre with one camera. Performance on other populations, cameras and image-quality levels, for example Messidor [35] or DDR [36], has not been tested.
- **Non-official split.** The random hold-out prevents direct comparison with the IDRiD challenge.
- **Weakly supervised lesion map.** The lesion-attention map has not been validated against lesion masks.
- **Training randomness.** Variance is reflected through five folds but not through repeated runs with different seeds.
- **Reference standard.** Grades are taken from the dataset providers; grader variability, which is substantial for DR [5], is not modelled.
- **Optimistic OOF estimates.** Checkpoints are selected on the same folds used for OOF predictions, so the cross-validation numbers (but not the test numbers) are slightly optimistic [20].

7.6 Ethical and clinical considerations

NITSCI is a research prototype, not a medical device. Both datasets are public and de-identified. Several issues must be addressed before any clinical use. First, models trained on data from one population can perform worse, or fail silently, on others; algorithmic bias in health care has had measurable consequences [80], so subgroup performance by age, sex, ethnicity and camera should be reported. Second, the operating point should be set together with clinicians to reach a target sensitivity rather than to maximise accuracy, since a false negative delays treatment while a false positive costs a specialist visit. Third, probabilities from deep networks are often miscalibrated [62]; calibration should be checked before a probability threshold is used for referral. Finally, prospective external validation reported according to TRIPOD+AI [63] is required, as autonomous DR systems in clinical use have shown [10].